

Tesla's China Decision: Build, Partner, or Wait? — Case Handout

Topic 1.3 PESTEL Factors · Read before class. Be ready to discuss.

This is a decision case. The clock stops in July 2018. Everything that happened afterward is deliberately hidden — do NOT look it up before the discussion. Your job is to make the decision yourself using only the information in this handout.

Background

Tesla, Inc. is an American electric vehicle (EV) company founded in Silicon Valley in 2003, with Elon Musk as CEO since 2008. Its Model S sedan (2012) and Model X SUV (2015) made Tesla the world's most admired EV brand. The Model 3, launched in 2017 at a starting price of \$35,000, was designed to bring electric cars to the mass market.

But in mid-2018 Tesla was under strain. The Fremont, California factory was struggling to hit Model 3 production targets — what Musk called "production hell" — and the company was burning through roughly \$1 billion per quarter. Tesla had never built a car outside the United States.

China had been Tesla's second-largest market since the first cars arrived in 2014. In 2017 Chinese customers bought about \$2 billion worth of Teslas — roughly 15–20% of the company's revenue. Every one of those cars was shipped from California, and every one paid China's import duties.

Now China was changing. The board had to decide: **how should Tesla compete in the most important car market on Earth?**

The Situation in July 2018 — A Board Briefing

1. The market

- China has been the world's largest car market since 2009: about **28 million vehicles** were sold there in 2017, compared with about 17 million in the United States.

- China also buys more electric cars than the rest of the world combined. Sales of "new energy vehicles" (NEVs — China's term for battery-electric and plug-in cars) hit **777,000 in 2017** and were heading for over 1.2 million in 2018. Government plans pointed toward NEVs making up about **20% of new-car sales by 2025**.
- Domestic rivals are moving fast: **BYD (比亚迪)**, partly owned by Warren Buffett's Berkshire Hathaway, was already China's top-selling NEV brand with cheaper models; well-funded startups **NIO (蔚来)**, **Xpeng (小鹏)**, and **Li Auto (理想)** were launching premium electric SUVs and sedans.
- Chinese EV buyers are younger and more urban than average; they expect large touchscreens, phone-as-key apps, and over-the-air software updates.

2. The rules for foreign automakers

- Since the 1980s, a foreign carmaker building in China could own **at most 50%** of a factory — it had to form a joint venture (JV) with a Chinese partner, limited to two partners. Every global automaker (GM, Volkswagen, Toyota) operated this way. Foreign firms long complained that JV partners expected **technology transfer** in return for market access.
- In **April 2018**, China's planning agency announced that the ownership cap would be removed for **NEV manufacturers in 2018** (commercial vehicles in 2020, ordinary passenger cars in 2022).
- A **trade war** began in mid-2018. China retaliated against U.S. tariffs with import duties that pushed the total tax on **U.S.-built cars to 40%** — while cars imported from Europe, Japan, and South Korea paid 15%.
- Government **subsidies** for NEV buyers (worth thousands of dollars per car in earlier years) and the exemption from the 10% vehicle purchase tax apply mainly to cars **built in China** and listed on government catalogs — imports largely miss out. Subsidies were already being phased down and were scheduled to end around 2022.
- High-definition **mapping and surveying** is restricted to licensed Chinese companies — a direct issue for self-driving features. Rules on vehicle-generated data (camera footage, location) were tightening.
- Foreign companies consistently rate **intellectual-property (IP) enforcement** in China as weak; Chinese JV partners have repeatedly launched their own brands after absorbing foreign technology.

3. The license-plate system

- In China's richest cities — Beijing (北京), Shanghai (上海), Shenzhen (深圳), Guangzhou (广州), Hangzhou (杭州) — buying a normal car requires winning a monthly lottery (Beijing's odds were **well below 1%**) or bidding at auction (a Shanghai plate averaged around **90,000 RMB**, roughly \$13,000–14,000).
- **New energy vehicles get free "green plates" immediately** in these cities. For many urban families, buying an EV is simply the only way to own a car at all.
- Severe winter **smog** (red alerts and school closures in recent years) made electric cars socially desirable and made the government's pro-EV policies widely popular.

4. Costs, capital, and the offer on the table

- Wages in Chinese auto plants are far below U.S. levels; the Yangtze River Delta around Shanghai is one of the densest auto-parts supply clusters in the world.
- A car built in China avoids ocean freight (roughly \$1,000–2,000 per vehicle), avoids the import duty, and becomes eligible for purchase-tax exemptions and green-plate treatment.
- A new car plant would cost **\$2 billion or more** — at a moment when Tesla is burning about \$1 billion per quarter. But Chinese banks and Shanghai officials had signaled support: a large, favorably priced site in **Lingang (临港)** and loans could be made available **for a wholly-owned plant**.
- The trade war could escalate or cool down. No one knew.

5. Technology and infrastructure

- By the end of 2018 China had about **300,000 public charging points** — roughly half of the world's total — built by the State Grid and private operators.
- Chinese battery maker **CATL (宁德时代)** was on its way to becoming the world's largest EV-battery producer, and China dominated lithium processing and battery-material supply chains.
- Chinese consumers live on super-apps (WeChat, Alipay) to a degree U.S. consumers do not; any car operating in China is expected to integrate with them.

6. Environment and resources

- Smog and climate policy drove a **dual-credit rule** (effective April 2018): large automakers must earn credits by selling NEVs, or buy credits from companies

that do. Every global carmaker selling in China now had to sell electric cars there.

- New 2018 rules made carmakers responsible for **battery traceability and recycling**.
- Counterpoint: China's electricity in 2018 was still roughly **two-thirds coal-fired** — an EV's environmental benefit depended on how the grid developed.

7. Tesla's own position

- **For it:** unmatched brand status in China (a Tesla was a status symbol), software and over-the-air-update leadership, and the Model 3 arriving at a price Chinese urban professionals could afford.
 - **Against it:** no local factory, no local partner, a thin service network, cars ineligible for most EV incentives, and top management fully absorbed by the Model 3 ramp at home.
 - **The window:** the Shanghai offer stood in mid-2018. People who do business in China know that policy windows can close as fast as they open.
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The Decision

At the July 2018 board meeting, four options were on the table:

- **Option A — Keep exporting from California.** No new capital, no technology exposure. Absorb or pass through the 40% tariff, and wait for the trade war to settle.
 - **Option B — Form a classic 50/50 joint venture** with a Chinese automaker, as every other foreign carmaker had done. A local partner handles government relations, land, and suppliers; Tesla shares ownership, profits — and technology.
 - **Option C — Build a wholly-owned Gigafactory in Shanghai.** A \$2 billion+ bet; the first wholly foreign-owned car plant in modern China. Full control, local costs, no tariff, incentive-eligible — and full exposure on intellectual property, data rules, and government relations.
 - **Option D — Wait and see.** Delay for a year until the trade war, subsidy cuts, and ownership rules become clearer — while BYD, NIO, and Xpeng keep growing and the Shanghai offer may cool.
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Questions

1. Respond to parts A, B, and C.

(A) Complete a PESTEL analysis for Tesla's China decision: for **each** of the six factors, identify one specific fact from this handout and mark it as an **opportunity** or a **risk** for Tesla.

Factor	One specific fact from the handout	Opportunity or risk?	How it acts on Tesla
Political			
Economic			
Social			
Technological			
Environmental			
Legal			

(B) For the political and legal factors you identified, explain the **direction** of each: is it pulling Tesla toward China or pushing it away?

(C) The 40% tariff "feels" economic, and the free green plates "feel" social. Classify each one by its **source**, and explain why.

2. Respond to parts A, B, and C.

(A) Identify the PESTEL factor the license-plate lottery system represents for Tesla (name the primary classification and one factor it overlaps with).

(B) Explain how the plate system changes consumer demand for electric cars in China's richest cities — and why it matters for the Model 3 specifically.

(C) If Beijing and Shanghai abolished all plate restrictions tomorrow, predict how EV demand would change. Which other factors would still support EV sales?

3. Respond to parts A, B, and C.

(A) Identify **two** benefits Option B (a 50/50 joint venture) would offer Tesla that Option C does not.

(B) Identify **two** specific risks Tesla would take on with Option C (the wholly-owned factory) that it would not face under Option A.

(C) Tesla's competitive advantage lies in its software, battery systems, and vehicle engineering. Evaluate which option best **protects** that advantage — and which option does the most to make Tesla **price-competitive** in China. Explain whether the same option does both.

4. Respond to parts A, B, and C.

(A) Choose the **three most important factors** for this specific decision and rank them.

(B) Justify your #1 factor: why does it matter more than the others for Tesla in July 2018?

(C) "A proper PESTEL analysis treats all six factors as equally important." Explain why this statement is wrong, using evidence from Tesla's situation.

5. Respond to parts A, B, and C.

(A) Recommend one option — **A, B, C, or D** (or a clearly sequenced combination) — for Tesla in July 2018.

(B) Justify your recommendation using **at least three specific factors** from your table. For each: name the factor, state its direction (opportunity or risk), and tie it directly to Tesla.

(C) State the single biggest risk of your recommended option and propose one concrete action Tesla could take to manage it.

6. (Challenge) Respond to parts A, B, and C.

PESTEL factors move — a factor that points one way in 2018 can reverse later.

(A) Identify one factor in this handout that is visibly changing as of July 2018.

(B) Explain how a reversal of that factor would change your recommendation.

(C) Recommend one **"trigger signal"** Tesla's board should watch — a specific event or data point that would tell management to speed up its China plans, or to pull back.

Keep this handout for reference during class discussion — and do not look up what Tesla actually decided until after the reveal.